Tilak Chowdary

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Summary

AI Engineer with around 1 year of hands-on experience building GenAI applications using LangChain, LangGraph, and RAG pipelines. Experienced in developing LLM-powered systems, retrieval architectures, and agentic workflows, with a focus on building practical AI solutions.

Technical Skills

Languages: Python, SQL
Generative AI: LLMs, LangChain, LangGraph, Advanced RAG, AI Agents, Prompt Engineering, Tool Calling
LLM APIs & Inference: OpenAI API, Groq API, Gemini API, Ollama, Hugging Face Transformers
Retrieval & Vector DBs: FAISS, ChromaDB, pgvector, BM25, Hybrid Retrieval, Reranking
Backend & Frameworks: FastAPI, Streamlit, REST APIs, SSE Streaming
Observability & Evaluation: LangSmith, RAGAS
ML & Data: Scikit-learn, Pandas, NumPy, NLP
Databases: MySQL, MongoDB
Tools & Platforms: Git, GitHub, Docker, VS Code, Jupyter Notebook
CS Fundamentals: Data Structures & Algorithms, OOP

Work Experience

AI Engineer Jan 2026 – Present
EduSync Solutions *Hyderabad, India*

- Functioning as an AI Engineer by conducting rigorous validation and stress-testing of AI-integrated ERP modules, specifically LLM-powered workflows, identifying critical edge cases and reducing pre-release defects.
- Partnered with cross-functional product and engineering teams to document technical requirements, verify AI feature behavior, and accelerate the deployment of intelligent workflow automation.
- Engineered an internal AI support-ticket triage prototype using Python, FastAPI, pgvector, and LLM APIs to automate categorization and reduce manual triage time.

AI & ML Intern Feb 2025 – May 2025
AiROBOSOFT Products and Services LLP *Bengaluru, India*

- Built a Python-based conversational chatbot using NLP preprocessing techniques for real-time user interactions.
- Designed chatbot dialogue flows and response handling logic, managing structured JSON-based inputs and outputs for conversational state.

Projects

Enterprise RAG Chatbot | *FastAPI, LangChain, FAISS, BM25, Groq, LangSmith, RAGAS* GitHub

- Architected a production-grade RAG system featuring hybrid retrieval (FAISS dense + BM25 sparse) and TF-IDF reranking, achieving 92% retrieval accuracy across 50 benchmark queries.
- Implemented SSE streaming, real-time source metadata citations, and multi-document querying capabilities, reducing p95 latency to 1.2 seconds.
- Integrated LangSmith observability and a RAGAS evaluation endpoint to automate retrieval and generation quality scoring, ensuring continuous system reliability.

AI Support Triage Agent | *Python, FastAPI, pgvector, OpenAI API, RAGAS*

- Built an AI agent that classifies support tickets by urgency and category using LLM structured outputs, retrieves similar past tickets via RAG, and drafts suggested responses.
- Implemented escalation guardrails with confidence scoring, routing low-confidence cases to human agents to prevent hallucinated responses.
- Designed a failure analysis dashboard with a 50-ticket golden dataset, tracking classification accuracy, retrieval precision, and hallucination rate via RAGAS and LLM-as-judge.

Education

Presidency University, Bangalore 2023 – 2025
Master of Computer Applications (MCA) | CGPA: 7.85/10

St. Joseph's Degree & PG College 2020 – 2023
Bachelor of Science, Computer Science | CGPA: 7.6/10

Achievements

- **IoT-Based Driver Alertness System** – Published and presented at ICISAS 2025 Conference, Dubai.
- Completed Data Analyst Certification – Frontline Media.